



Assessing Variation in Built Environment Factors and Heat Adaptation in San Francisco

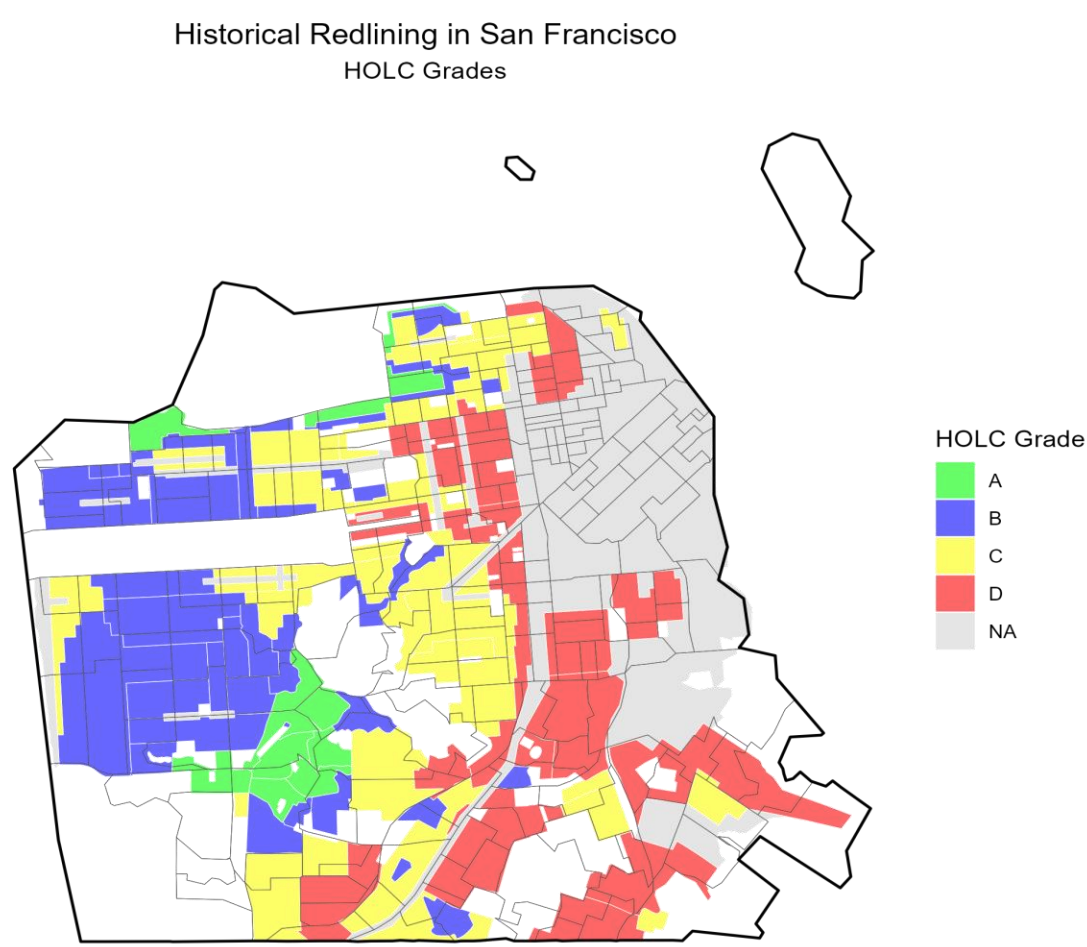
Alani Leonard¹, Matt Holmes², Melina Rubio^{2 3}, Marcelina Lobato³,
Amethyst Blumenshine², Mahal Leia Bilaoen BS³, Nancy Carmona PhD²

Department of Biology¹, Department of Public Health², The School of Environment³



Introduction

- **Rising extreme heat conditions** are an increasing threat to **urban residential health**.
- **Cooling investments** reduce **heat-related negative health outcomes**.
- Higher **Social Vulnerability Index (SVI)** and **lower historical redlining grades** are linked to **lower cooling-related infrastructure investments**



Methods

- Using Steadman's formula for apparent temperature to calculate what it feels like for people during a hot day
- Heatwaves defined as 2+ consecutive days above the 95th percentile of historical (1982-2012) normal apparent temperature
- Analyses included Wilcoxon tests, Spearman correlations, and logistic regression.

$$A = -1.3 + 0.92T + 2.2e$$

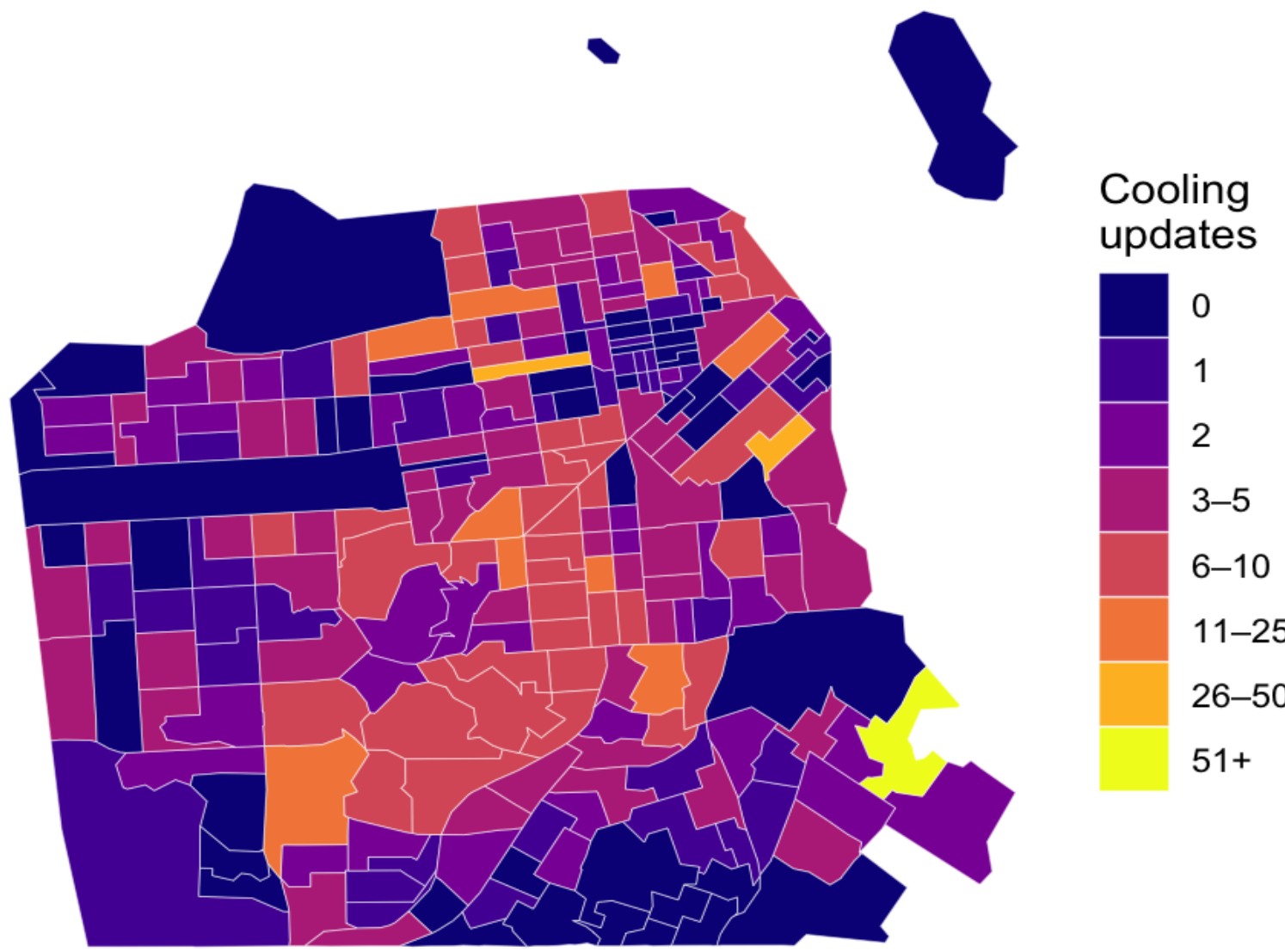
- A = Apparent Temperature
- T = Temperature (°C)
- e = water pressure (kilopascals)

Data source	Key features used	Years	Analysis unit
Daymet	Apparent temperature; heatwave events	1982–2022	Census tract-year
SF Building Permits + Assessor Tax Rolls	Cooling-related permits; year built; building age	2013–2022	Census tract-year
SF Tree Inventory Planting Data	Trees planted (proxy for greening investment)	2013–2022	Census tract-year
CDC Social Vulnerability Index (SVI)	Overall SVI + four domain scores	2022	Census tract
Mapping Inequality (Home Owner's Loan Corporation)	Historical HOLC redlining grade (A–D); dominant tract grade	Historical (1930s)	Census tract

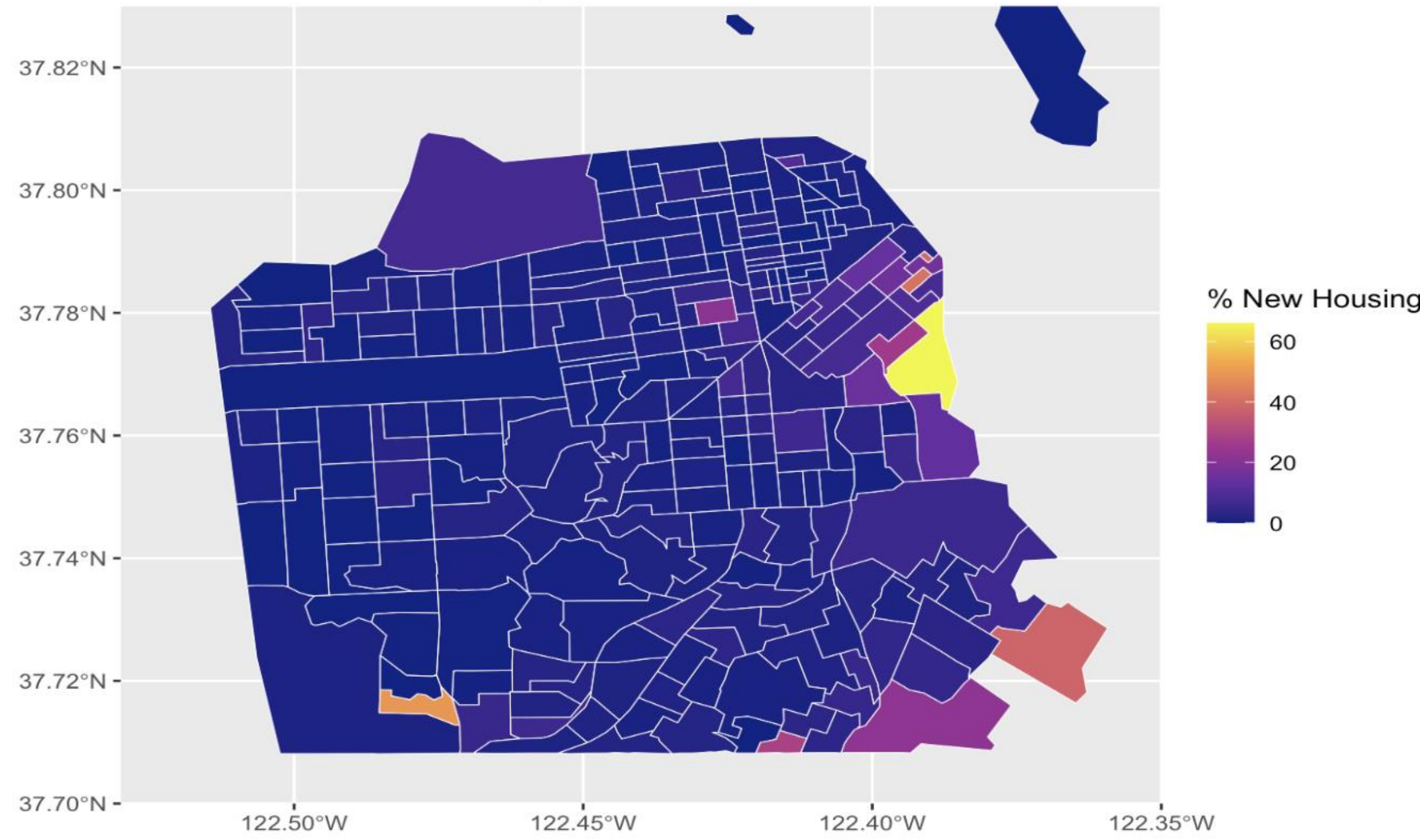
Results

- Grade A tracts had the highest median tree planting activity, with **8 trees per tract-year** compared with **3–4 trees** in Grades B–D.
- Higher-heat tract-years had more tree planting and cooling-related building updates than lower-heat tract-years ($p < 0.001$ and $p = 0.002$).
- Compared with Grade A tracts, Grades B–D had **47%–56% lower adjusted odds** of cooling-related building updates.
- The Mission had the most trees planted, followed by Bayview Hunters Point and Sunset/Parkside.

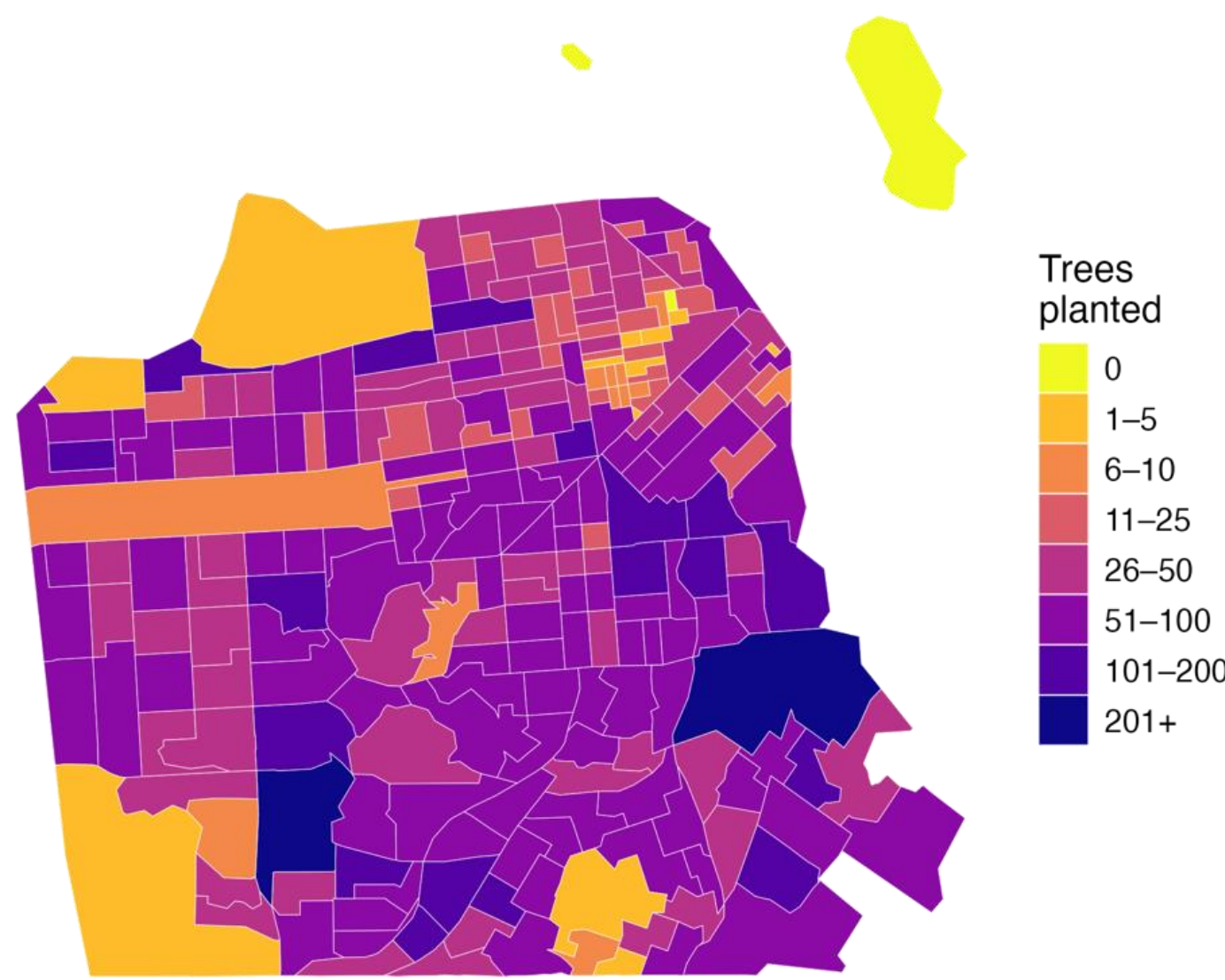
Residential Cooling Updates by Census Tract
Total residential cooling-related permits, 2013–2022



Percent of Newer Housing by Census Tract



Trees Planted by Census Tract
Total trees planted, 2013–2022



Characteristic	N	A: Best N = 67 ¹	B: Still desirable N = 336 ¹	C: Definitely declining N = 600 ¹	D: Hazardous N = 610 ¹	p- value ²
Social Vulnerability Index	1,613	0.10 (0.05, 0.21)	0.37 (0.26, 0.49)	0.17 (0.06, 0.43)	0.45 (0.26, 0.61)	<0.001
2-day heatwave events	1,613	8.0 (4.0, 10.0)	7.0 (4.0, 11.0)	7.0 (3.0, 10.0)	7.0 (4.0, 9.0)	0.5
Cooling/air updates	1,613	0.00 (0.00, 1.00)	0.00 (0.00, 0.00)	0.00 (0.00, 1.00)	0.00 (0.00, 0.00)	<0.001
Trees planted	1,613	8 (3, 17)	3 (1, 7)	4 (1, 7)	3 (1, 8)	<0.001
Median building age	1,613	99 (91, 108)	90 (83, 97)	102 (95, 114)	105 (79, 114)	<0.001

¹ Median (Q1, Q3)

² Kruskal-Wallis rank sum test

Conclusion and Implications

- Heat-adaptation resources were more common in hotter census tract-years, suggesting **some investment is reaching areas with greater heat exposure**.
- Cooling-related updates were less likely in historically lower-graded HOLC neighborhoods, indicating **persistent geographic inequities**.
- Future investments should be equitable and prioritize neighborhoods with high heat exposure, greater social vulnerability, and fewer existing cooling resources.

Poster pdf / Citations qr code

PINC Summer Program
Promoting Inclusivity in Computing

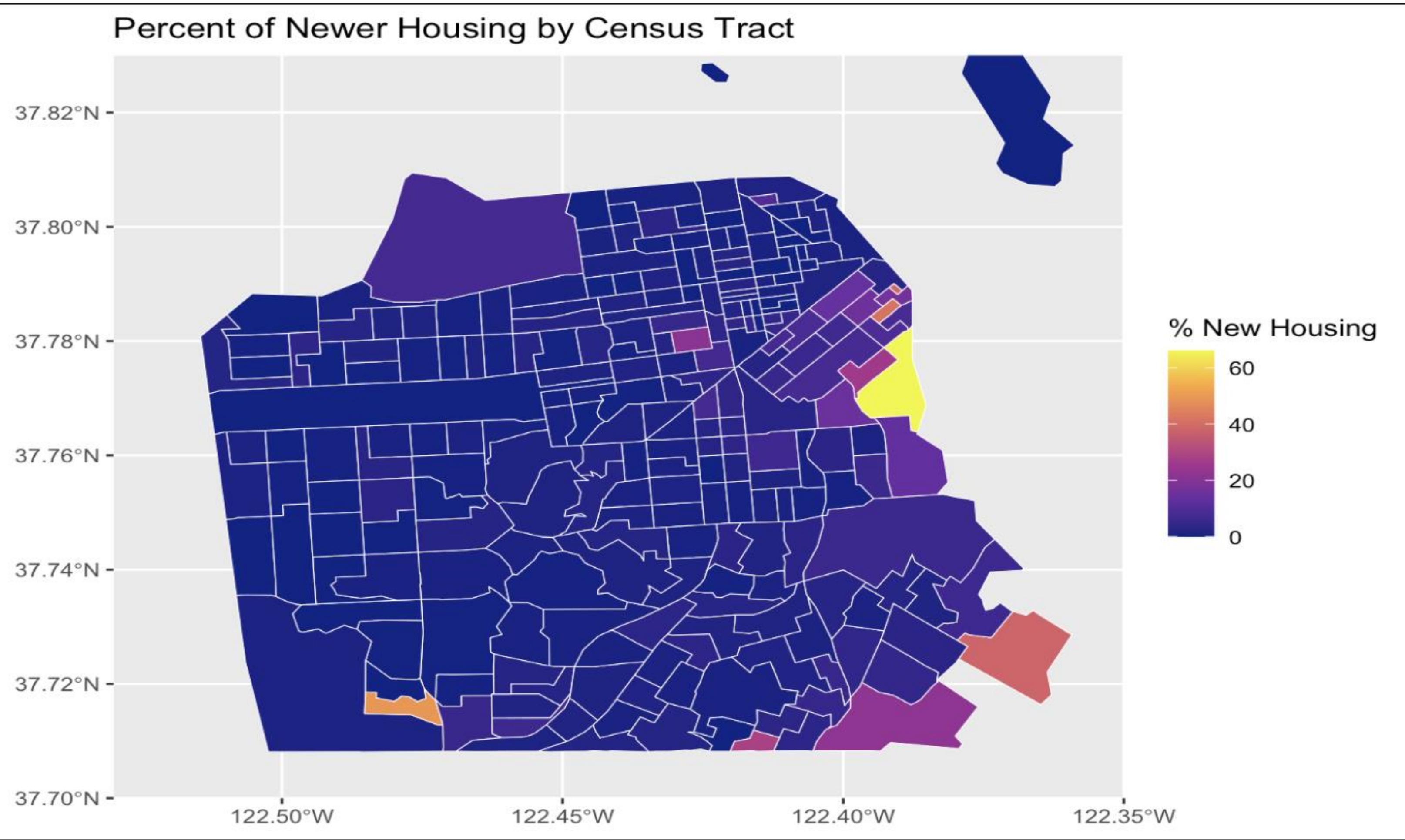
with funding from
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Citations (ACS) - dumping everything in first, reorganize + add numbering , comment the title/ author if you know where the citation needs to go – we will add in superscripts once we have all of our sources

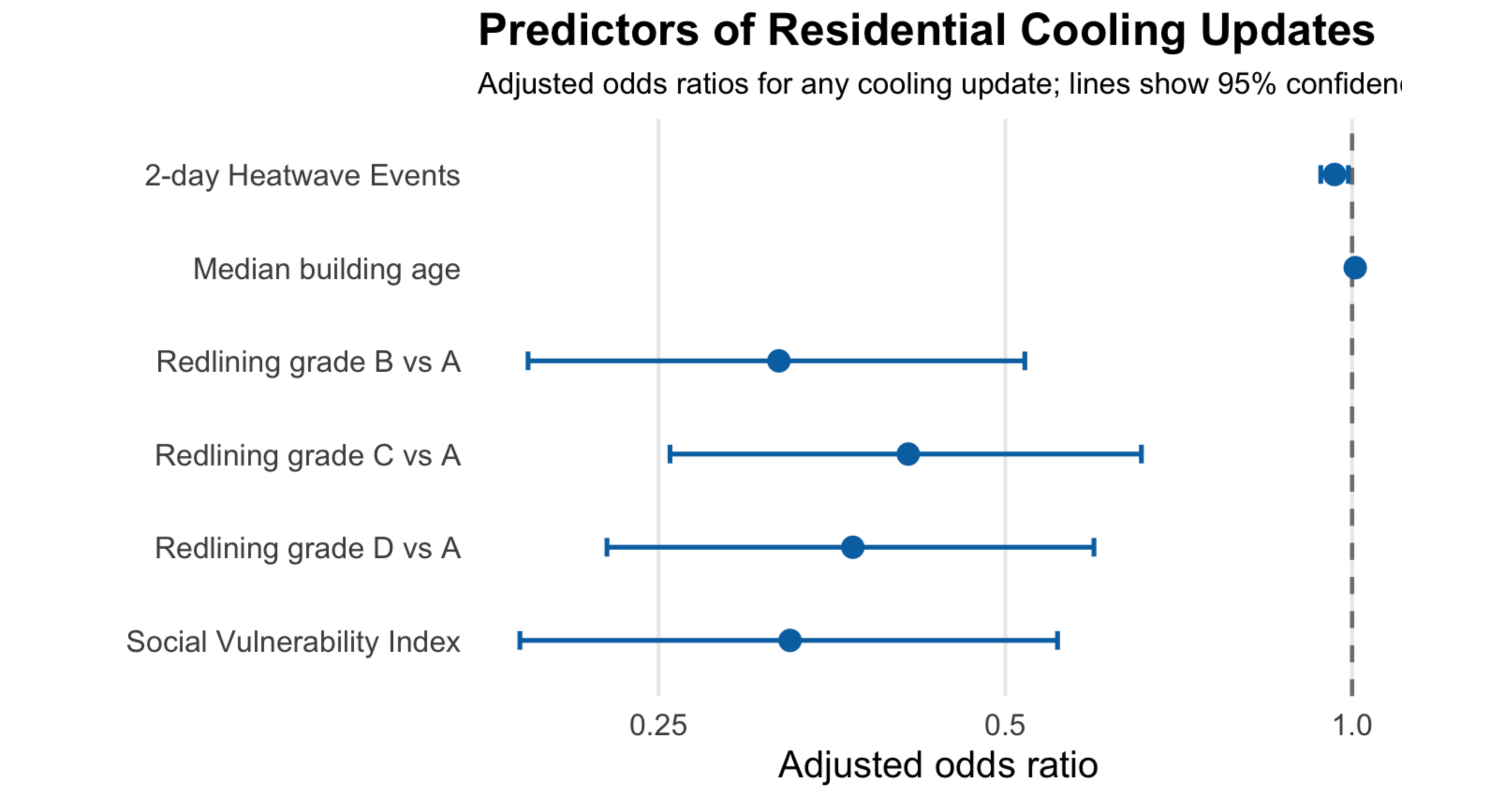
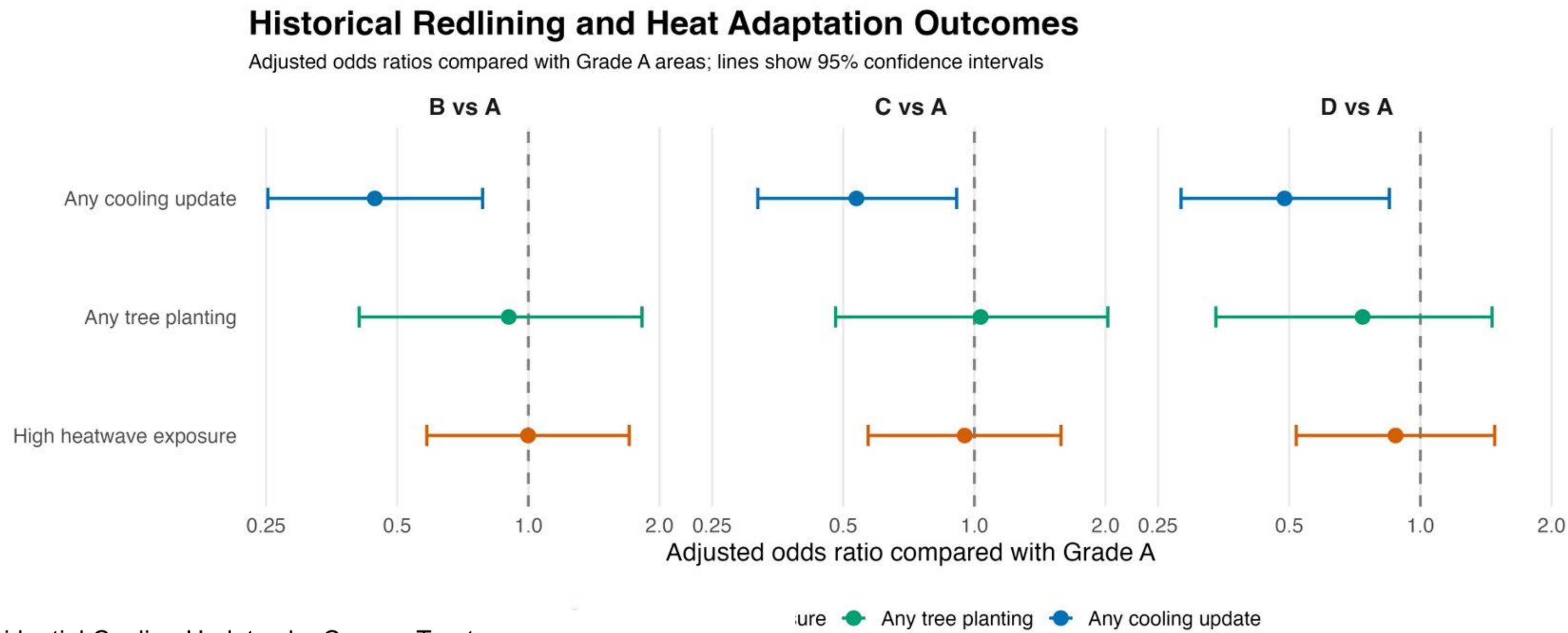
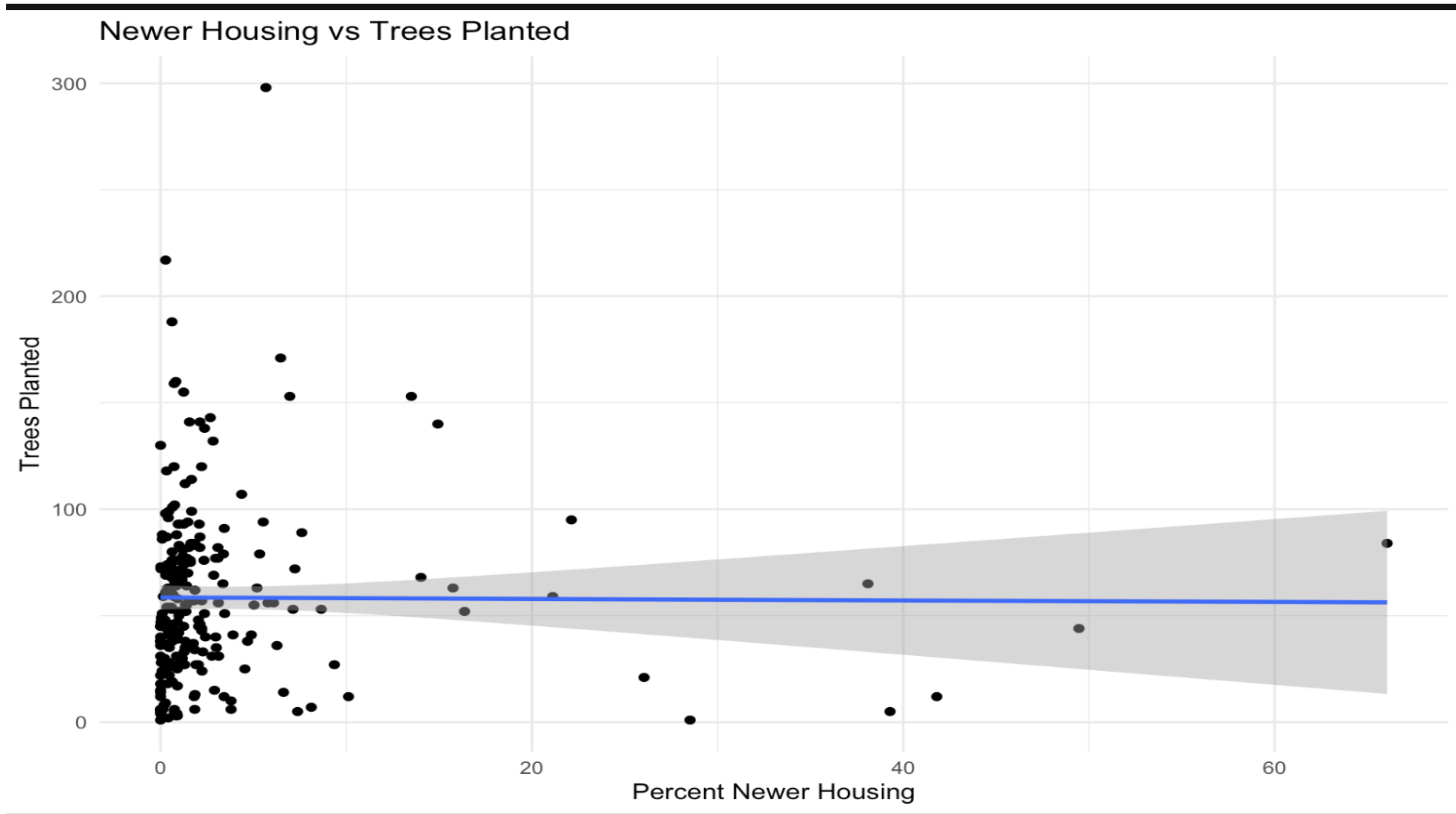
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 7. *Heat and health*. <https://www.who.int/news-room/fact-sheets/detail/climate-change-heat-and-health> (accessed 2025-10-08).
 8. Add in citation for redlining / holc grade [Mapping Inequality](#)
 9. DataSF. *Building Permits Before January 1, 2013*. <https://data/sfgov.org>
 10. DataSF. *Building Permits Filed on or After January 1, 2013*. <https://data/sfgov.org>
 11. DataSF. *Street Tree List*. <https://data.sfgov.org>
- Tax roll data (?) check through code
–[Assessor Historical Secured Property Tax Rolls | DataSF](#)



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